Assignment 3

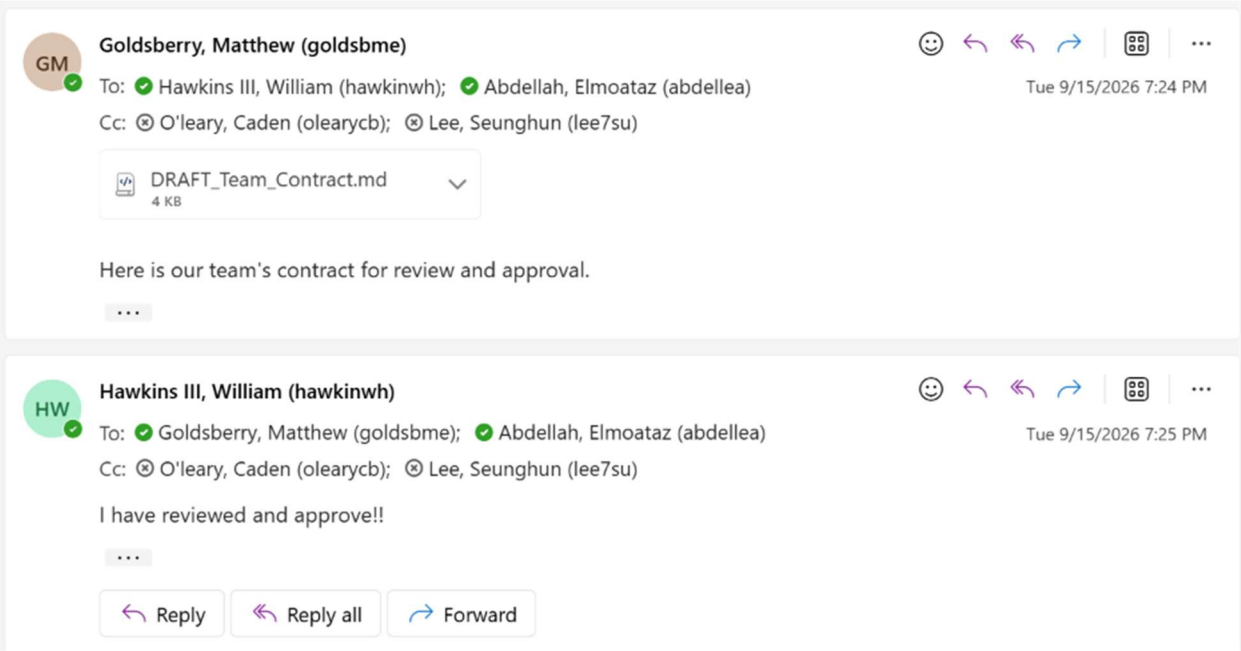
Caden O'Leary, Elmoataz Abdellah, Matthew Goldsberry, Seunghun Lee, Kristen Griessman

Advisor

Advisor: Our advisor is William Hawkins III. He works in the computer science department. His email is hawkinwh@ucmail.uc.edu.

Reasons for selection: Dr Hawkins has already worked with the CubeCats organization for previous projects (LEOPARDSat-1). He has also done many research papers on low level development.

Email Approval:



Team Contract

- **Project:** HABSat-1 Flight Software
- **Project Due Date:** End of April 2027
- **Project Members:** Caden O'Leary, Elmoataz Abdellah, Matthew Goldsberry, Seunghun Lee

Roles & Responsibilities

Member	Role	Responsibility
Matthew Goldsberry	OBC Team Lead (Coordinator)	Responsible for leading the OBC team (including members of the OBC team outside of this senior design group in the club) to make sure we are meeting project deliverable deadlines and requirements for both senior design and HABSat-1's project timeline.
Caden O'Leary	OBC Documentation Owner	Maintains the design report, notebook consistency, and submission formatting.
Elmoataz Abdellah	OBC QA/ Test Owner	Owns test planning and validation against requirements.
Seunghun Lee	OBC Developer / Architect	Technical direction and key design decisions for the OBC software and hardware architecture.
Kristen Griessman	OBC Security Owner	Responsible for ensuring the security of the OBC software and hardware architecture.

Communication Ground Rules

Primary communication will occur in the dedicated project Discord server. Official documents, meeting minutes, and code will be centralized in GitHub and Google Drive.

Meeting Cadence: Weekly meetings amongst just the senior design team will occur every Wednesday immediately following the senior design class.

Lab Sessions (optional, but strongly encouraged):

- Tuesdays: 6PM-8PM
- Sundays: 12PM-1PM

Advisor Meeting: 9/15/2026

Response Expectations: All members will acknowledge Discord pings within 1 day of receiving them, barring planned absences or illness.

Attendance Policy: If a team member cannot attend a scheduled meeting or lab session, they must provide notice at least 6 hours in advance.

Decision-Making Process

- **Consensus-Driven Approach:** All major design and architectural decisions go through a proposal and discussion, and are decided via team consensus.
- **Fallback and/or Tie-Breaking:** If consensus cannot be reached, a majority vote (3-to-1) decides the path forward. In the event of a tie (2-to-2), the tie-breaking vote will be handed to the faculty advisor to make the final, executive decision.
- **Technical (Software) Changes:** Code will not be merged directly into the main branch. Every feature/issue will have the requirement of one approval before the Pull Request can be merged.

Accountability & Conflict Resolution

Ownership: Every member will take personal ownership of their work. If a blocker or external factor pops up that risks making a deadline or goal, the owner of that must notify the team immediately so tasks can properly be reallocated.

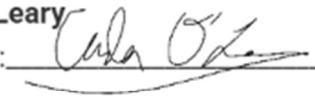
Conflict Resolution:

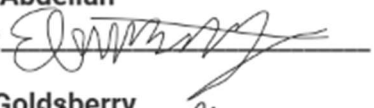
- **Direct Discussion:** The conflicting members will directly address the issue to try to find an actionable compromise.
- **Team Mediation:** If unresolved, the issue will be escalated to a team discussion to try to reach an internal resolution. Resolution should be focused on meeting project deliverables.
- **Advisor:** If the team cannot resolve it, we will then escalate to the faculty advisor to try to resolve the conflict.

- **Course Coordinator:** If the team and advisor cannot resolve the issue, the final stage is bringing the conflict to the course coordinator for resolution.

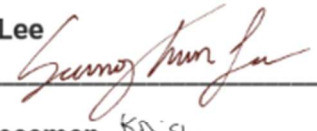
Signatures

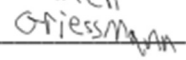
By signing below, each team member acknowledges that they have read, understood, and agreed to uphold all terms set forth in this contract throughout the duration of the HABSat-1 Flight Software project.

Caden O'Leary
Signature:  Date: 9/16/26

Elmoataz Abdellah
Signature:  Date: 9/16/26

Matthew Goldsberry
Signature:  Date: 9/16/26

Seunghun Lee
Signature:  Date: 9/16/26

Kristen Griessman
Signature:  Date: 9/16/26

As the faculty advisor to this project, I acknowledge that I have reviewed and endorse this team contract.

Will Hawkins III
Signature: [Approved Via Email on 9/15/2026] Date: 9/15/2026

Project Constraints

Economic

- Due to the financial limitations of the CubeCats organization, we are limited to their already purchased hardware and budget for this project.
- Because of this, we will focus on using open-source tooling and will not use proprietary software.

Professional

- This project must undergo a preliminary design review by professionals in the aerospace industry as part of the overarching project lifecycle.
- Since we are operating within an organization with the University of Cincinnati, our actions and work will reflect the public opinion of the school and the CubeCats organization.
- The flight software shall follow established aerospace software engineering practices for requirements management, version control, testing, documentation, and verification.

Legal

- Comply with all US regulations and procedures relating to export control and research security.
- Comply with all US regulations and procedures for the utilization of radio frequencies.

Security

- Payload data must be encrypted on downlinks to prevent foreign actors from accessing sensitive data.
- Critical spacecraft commands shall require additional validation before execution to reduce the risk of accidental or malicious commands.
- The flight software shall isolate faults within individual subsystems to prevent a failure in one subsystem from unnecessarily compromising the operation of the OBC or other spacecraft subsystems.